

Theory of Programming Languages

4. Natural Deduction

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Logic

- What is logic? A tool for reasoning about truths
- Why logic for computer scientists?
- Verification: reasoning about computation (CS424, CS524)
 - “Does this program accept an array of integers and produce a sorted array?”
 - “Does this program access an unallocated memory?”
 - “Does this function always halt?”
- Language design: intrinsic connections with programming languages (this course)
- Fundamental question: “What does ‘true’ mean?”

Intuitionistic Logic (1) 직관논리

- Introduced by Arend Heyting (1928)
- Central question: “What is the meaning of ‘ P is true’?”
 - Classical (고전논리): “When is P true?”, “ P holds in all possible cases”
 - Intuitionistic (직관논리): “What is a proof of P ?”, “ P is directly probable”
- Key difference: law of excluded middle (LEM, 둘중하나), $P \vee \neg P$
 - Classical: accept LEM
 - Intuitionistic: reject LEM
- Also known as constructive logic

Intuitionistic Logic (2)

- Example: $\exists x, y \in \mathbb{I} . x^y \in \mathbb{Q}$?
 - Classical: Yes
 - Intuitionistic: Don't know
- Analogy
 - Schrödinger's cat
 - “Whereof one cannot speak, thereof one must be silent” - Wittgenstein

Natural Deduction

- Introduced by Gerhard Gentzen (1935)
- System for logical reasoning
- Idea:
 - Proof = tree-like structure (possibly multiple hypotheses & single conclusion)
 - Proof by symmetric procedures, i.e., introduction & elimination of connectives (연산자)
- Proof rule (or inference rule)

$$\frac{P_1 \ \dots \ P_n \text{ premise(s)}}{P \text{ conclusion}}$$

$$\frac{\text{human}(x) \implies \text{mortal}(x) \quad \text{human}(s)}{\text{mortal}(s)}$$

Example: Propositional logic

- Syntax: $A \rightarrow P \mid A \wedge A \mid A \rightarrow A \mid A \vee A \mid \top \mid \perp \mid \neg A$
- Γ : a set of hypotheses
- Proof rules: set of introduction and elimination rules
 - Introduction rule: introduce connectives
 - Elimination rule: use connectives

ND for Propositional logic (1)

- Hypothesis

$$\frac{A \in \Gamma}{\Gamma \vdash A} \text{Hyp}$$

- Implication

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightarrow B} \rightarrow I \quad \frac{\Gamma \vdash A \rightarrow B \quad \Gamma \vdash A}{\Gamma \vdash B} \rightarrow E$$

- Conjunction

$$\frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \wedge I \quad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \wedge EL \quad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B} \wedge ER$$

ND for Propositional logic (2)

- Disjunction

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B} \vee\text{IL} \quad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B} \vee\text{IR} \quad \frac{\Gamma \vdash A \vee B \quad \Gamma, A \vdash C \quad \Gamma, B \vdash C}{\Gamma \vdash C} \vee\text{E}$$

- True and False

$$\frac{}{\Gamma \vdash \top} \top\text{I} \quad \frac{\Gamma \vdash \perp}{\Gamma \vdash C} \perp\text{E}$$

- Negation (syntactic sugar: $\neg A \equiv A \rightarrow \perp$)

$$\frac{\Gamma, A \vdash \perp}{\Gamma \vdash \neg A} \neg\text{I} \quad \frac{\Gamma \vdash \neg A \quad \Gamma \vdash A}{\Gamma \vdash \perp} \neg\text{E}$$

ND for Propositional logic (3)

- Classical logic: intuitionistic logic + one of the followings (one can derive the others)

$$\frac{}{\Gamma \vdash A \vee \neg A} \text{EM} \quad \frac{}{\Gamma \vdash \neg\neg A \rightarrow A} \text{DNE} \quad \frac{}{\Gamma \vdash ((A \rightarrow B) \rightarrow A) \rightarrow A} \text{Peirce}$$

Exercise

- Prove $A \rightarrow (B \rightarrow A)$

$$\frac{\frac{\frac{}{A, B \vdash A} \text{Hyp}}{A \vdash B \rightarrow A} \rightarrow\text{I}}{\vdash A \rightarrow (B \rightarrow A)} \rightarrow\text{I}$$

Exercise

- Prove $(A \rightarrow B) \wedge (A \rightarrow C) \rightarrow (A \rightarrow B \wedge C)$

$$\frac{\dots}{\vdash (A \rightarrow B) \wedge (A \rightarrow C) \rightarrow (A \rightarrow B \wedge C)}$$

Exercise

- Prove $((A \rightarrow B) \rightarrow A) \rightarrow A$

Excercise

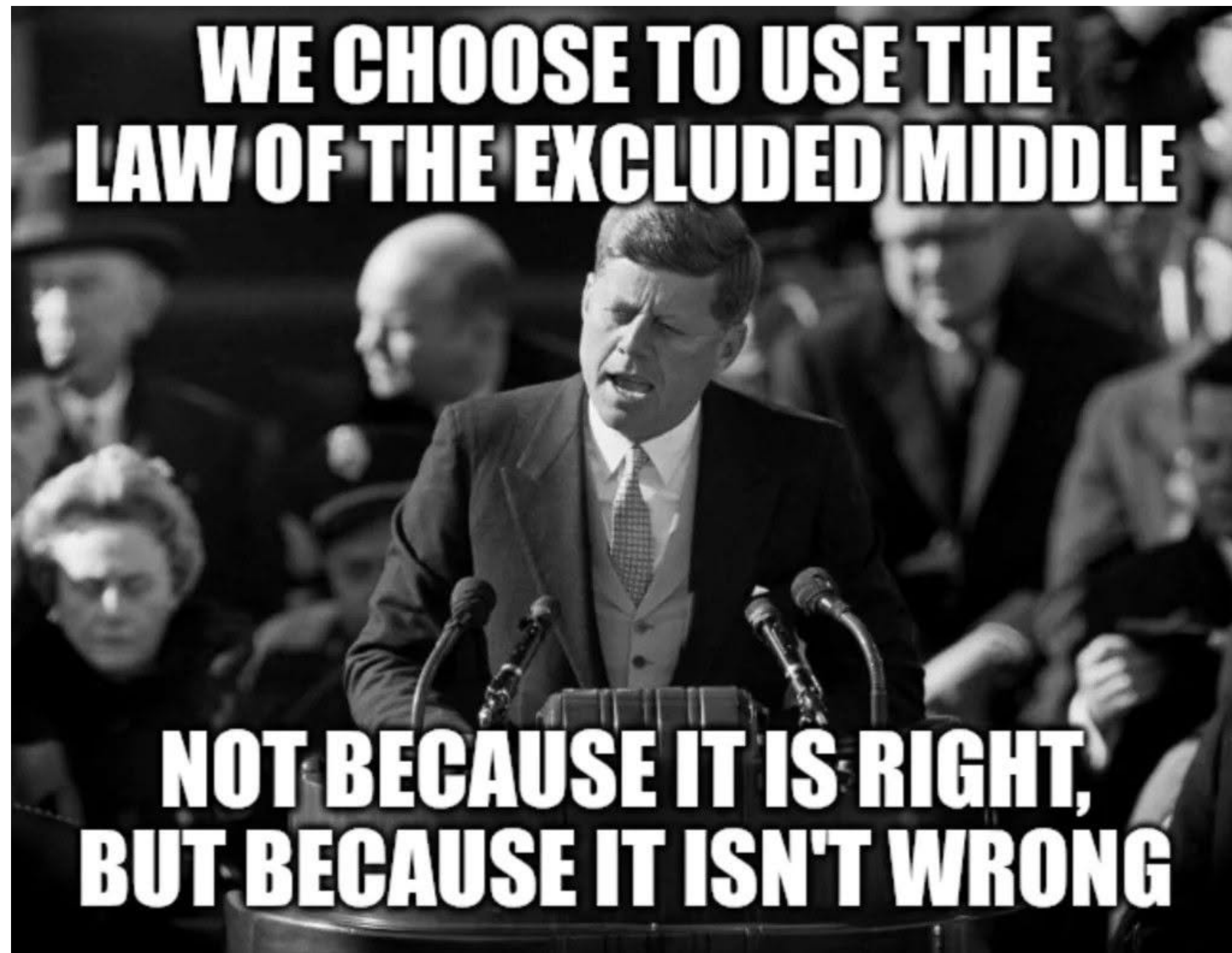
- Goal: $\sqrt{2}$ is not rational
- Proof*: Assume $\sqrt{2}$ is rational. ... It is a contradiction. So $\sqrt{2}$ is not rational
- Is this intuitionistically valid proof?

*https://en.wikipedia.org/wiki/Square_root_of_2#Proof_by_infinite_descent

Exercise

- Goal: If in a triangle two angles equal one another, then the sides opposite the equal angles also equal one another.
- Proof*: Assume the sides are not equal. ... It is a contradiction. So they are equal.
- Is this intuitionistically valid proof?

*<http://aleph0.clarku.edu/~djoyce/elements/bookI/propI6.html>



*Mathematical Mathematics Memes, Facebook

Summary

- Natural deduction: a systematic way to prove logical propositions
 - Key idea: introduction and elimination rules
- Two camps of logic
 - Classical: “When is P true?”, “ P holds in all possible cases”
 - Intuitionistic: “What is a proof of P ?”, “ P is directly probable”
 - Classical = Intuitionistic + LEM